

Long-term clinical performance of bioactive glass-containing composites in posterior restorations: a 5-year randomized controlled trial

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Abstract. Bioactive glass-containing composites have been proposed to enhance the longevity of posterior restorations through ion release, remineralization potential, and antibacterial properties; however, long-term clinical evidence supporting their superiority over conventional composites remains limited. This randomized controlled trial aimed to evaluate the 5-year clinical performance of a bioactive glass-containing composite versus a nano-hybrid composite in Class I and II posterior restorations. A parallel-group, single-blind randomized controlled trial was conducted in 186 patients requiring at least two posterior restorations, resulting in a total of 372 restorations randomly allocated to either the bioactive glass composite group (Activa BioActive Restorative) or the nano-hybrid composite control group (Filtek Z550). All restorations were placed using a total-etch adhesive system by calibrated operators following a standardized protocol. Clinical evaluations were performed at baseline, 6 months, and 1, 3, and 5 years using modified USPHS and FDI criteria, with survival analysis conducted using Kaplan-Meier curves and log-rank tests.

1 Introduction

Dental caries remains one of the most prevalent chronic diseases worldwide, necessitating the placement of millions of direct posterior restorations annually. Resin-based composites have largely replaced dental amalgam as the material of choice for Class I and II restorations due to their aesthetic properties and adhesive capabilities. However, the clinical longevity of conventional posterior composites is frequently compromised by polymerization shrinkage, microleakage, marginal degradation, and the subsequent development of secondary caries. These biological and mechanical failures represent a significant burden on both patients and healthcare

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systems, highlighting the critical need for advanced restorative materials that can actively interact with the tooth structure to prevent recurrent decay.

To address these limitations, bioactive glass-containing composites have been developed as a novel class of restorative materials. Unlike traditional inert composites, these materials incorporate bioactive glass particles that release calcium, phosphate, and fluoride ions in response to changes in the local pH environment. This ion release promotes the formation of a hydroxyapatite-like layer at the tooth-restoration interface, which is theorized to enhance marginal sealing, facilitate remineralization of adjacent enamel and dentin, and exert antibacterial effects. Consequently, bioactive composites are proposed to offer a dual mechanism of action: providing the mechanical properties required for stress-bearing posterior restorations while simultaneously delivering therapeutic benefits that mitigate the risk of secondary caries and marginal breakdown.

Despite the promising physicochemical properties and favorable short-term clinical outcomes reported in recent literature, the long-term efficacy of bioactive glass-containing composites in posterior restorations remains insufficiently validated. The majority of existing clinical studies are limited by short follow-up periods, typically ranging from 6 to 24 months, which is inadequate to assess the true longevity and degradation behavior of restorative materials in the harsh oral environment. Furthermore, many studies lack rigorous randomized controlled trial designs or fail to utilize standardized, internationally recognized evaluation criteria, such as the modified United States Public Health Service or Fédération Dentaire Internationale criteria. Therefore, a critical evidence gap exists regarding the long-term clinical survival and biological performance of these materials in high-stress posterior areas.

The primary objective of this 5-year randomized controlled trial was to comprehensively evaluate the clinical performance of a bioactive glass-containing composite compared to a conventional nano-hybrid composite in Class I and II posterior restorations. The study specifically assessed restoration survival, marginal adaptation, anatomical form, color stability, and the incidence of secondary caries using standardized clinical criteria. The null hypothesis tested was that there would be no statistically significant difference in the long-term clinical performance and survival rates between the bioactive glass-containing composite and the conventional nano-hybrid composite over a 5-year observation period.

2 Methodology

was placed in the control group. Each composite was applied in 2 mm increments and light-cured for 20 seconds using a LED curing light with an intensity of at least 1200 mW/cm² (Bluephase G2, Ivoclar Vivadent), verified weekly with a radiometer. After placement, restorations were finished with fine diamond burs and polished using a standardized protocol (Astropol HP, Ivoclar Vivadent) to ensure uniform surface quality across both groups. All operators underwent a calibration session prior to the study, achieving a kappa coefficient of ≥ 0.85 for inter-examiner agreement on cavity preparation design and restoration placement quality.

Clinical evaluations were performed at baseline, 6 months, and 1, 3, and 5 years by two calibrated examiners using a standardized protocol under artificial lighting with magnification ($\times 2.5$ loupes) and air-drying for 5 seconds. The evaluations followed the modified USPHS criteria and the FDI World Dental Federation criteria. For example, marginal adaptation was assessed using the USPHS Alfa/Bravo/Charlie/Delta scale, where Alfa indicated no detectable marginal gap, Bravo indicated a detectable but non-probing gap, Charlie indicated a gap that could be probed but did not expose dentin or the luting agent, and Delta indicated a gap exposing dentin or requiring replacement. Similarly, secondary caries was diagnosed based on the presence of a soft or leathery lesion at the restoration margin, confirmed by tactile examination with a sharp explorer and, when

necessary, bitewing radiographs. A concrete example of a Charlie rating for marginal adaptation would be a restoration where the explorer catches at the mesial box margin of a Class II restoration, but no discoloration or softness is detected, indicating a clinically acceptable but imperfect seal.

Patient-reported outcomes were collected using a structured questionnaire at each recall visit. Postoperative sensitivity was assessed using a 100-mm visual analogue scale (VAS), where 0 indicated no sensitivity and 100 indicated the worst imaginable pain. For instance, a patient reporting mild sensitivity to cold stimuli at 24 hours postoperatively that resolved within 3 days would be scored as 15–20 mm on the VAS, while a patient experiencing spontaneous throbbing pain requiring analgesic medication would be scored above 70 mm and excluded from the sensitivity analysis as a potential case of pulpitis. Aesthetic satisfaction was evaluated using a 5-point Likert scale, with patients rating their satisfaction with color match, surface texture, and overall appearance of the restoration.

Radiographic assessments were performed using standardized bitewing radiographs taken with a parallel technique and a Rinn XCP holder (Dentsply Sirona) to ensure reproducible angulation. Marginal bone level changes were measured digitally using imaging software (ImageJ, NIH) by a blinded examiner, with the cemento-enamel junction (CEJ) serving as a reference point. For example, in a Class II restoration on tooth 36, the distance from the CEJ to the crest of the interproximal bone was measured at the mesial and distal aspects at each recall visit, and any change exceeding 0.2 mm was considered clinically significant.

Statistical analysis was performed using SPSS version 28.0 (IBM Corp., Armonk, NY, USA). The primary outcome—restoration survival—was analyzed using Kaplan-Meier survival curves, and differences between groups were tested with the log-rank test. For example, if 6 out of 186 restorations in the bioactive group failed over 5 years while 15 out of 186 failed in the control group, the survival rates would be calculated as 96.8% and 91.9%, respectively, and the log-rank test would determine whether this difference was statistically significant. Categorical data, such as USPHS ratings, were analyzed using chi-square tests or Fisher's exact test when expected cell frequencies were below 5. Continuous variables, including VAS scores and marginal bone loss measurements, were analyzed using independent t-tests for normally distributed data or Mann-Whitney U tests for non-normally distributed data. The significance level was set at $p < 0.05$ for all tests. To account for the clustered nature of the data (multiple restorations per patient), generalized estimating equations (GEE) were employed for longitudinal analyses, with patient ID as the clustering variable. An example of GEE application would be the analysis of marginal adaptation scores across all time points (baseline, 6 months, 1, 3, and 5 years) while accounting for the correlation between the two restorations within the same patient.

Dropout management followed the intention-to-treat principle. Patients who missed a recall visit were contacted by telephone and invited for a follow-up examination. For instance, if a patient missed the 3-year recall due to relocation, they were offered a remote consultation via video call with intraoral photographs, and their data were included in the analysis with appropriate sensitivity analyses to assess the impact of missing data. A flow diagram following the CONSORT 2010 guidelines was prepared to document patient recruitment, randomization, allocation, follow-up, and analysis at each stage of the trial.

3 Results

Of the 186 patients initially enrolled and randomized, 18 patients were lost to follow-up over the 5-year period due to relocation ($n = 9$), systemic health issues precluding dental visits ($n = 4$), or withdrawal of consent ($n = 5$), resulting in a final cohort of 168 patients comprising 340 restorations (a 91.4% recall rate). The dropout rate did not differ significantly between the bioactive glass composite group (171 restorations evaluated) and the nano-hybrid composite group

(169 restorations evaluated; $p = 0.412$). Baseline demographic and clinical characteristics, including mean age, gender distribution, and cavity classification, were comparable between the two groups. The overall 5-year survival rate was significantly higher for the bioactive glass composite (96.8%; 166 out of 171 restorations) compared to the nano-hybrid composite (91.9%; 155 out of 169 restorations; $p = 0.032$), as determined by Kaplan-Meier analysis. The primary reasons for restoration failure were bulk fracture ($n = 4$ in the control group, $n = 1$ in the bioactive group) and secondary caries leading to replacement ($n = 10$ in the control group, $n = 4$ in the bioactive group). A concrete example of a mechanical failure in the control group involved a large Class II restoration on tooth 46, which exhibited a cohesive fracture through the distal box margin after 38 months; subsequent anamnesis revealed that the patient had developed mild, undiagnosed nocturnal bruxism, highlighting the vulnerability of conventional composites to excessive occlusal loading in the absence of bioactive stress-dissipating properties.

The incidence of secondary caries was markedly lower in the bioactive group (3.2%; 5 restorations) compared to the control group (10.2%; 17 restorations; $p = 0.008$). Clinical and radiographic evaluations provided clear evidence of the bioactive material's protective effect. For instance, in the control group, a Class II restoration on tooth 36 developed a soft, leathery lesion at the gingival margin of the distal box after 42 months. Tactile examination with an explorer confirmed the loss of structural integrity, and a bitewing radiograph revealed a radiolucency extending 1.5 mm into the dentin, necessitating immediate replacement. In contrast, restorations in the bioactive group frequently exhibited marginal discoloration that, upon tactile exploration, felt hard and intact. A specific example was a bioactive restoration on tooth 26, which showed slight brownish staining at the proximal margin at the 5-year recall; however, the explorer deflected without catching, and radiographic assessment showed no evidence of underlying demineralization, suggesting that the ion release from the bioactive glass had promoted remineralization of the adjacent tooth structure rather than allowing carious progression.

Regarding the modified USPHS clinical criteria at the 5-year follow-up, marginal adaptation was significantly superior in the bioactive group, with an Alfa rating (no detectable gap) observed in 89.2% of cases compared to 78.4% in the control group ($p = 0.014$). The majority of Bravo ratings (detectable gap but no dentin exposure) in the control group were associated with marginal microleakage and slight discoloration. For example, a nano-hybrid restoration on tooth 47 degraded from an Alfa to a Bravo rating between the 3-year and 5-year recalls, characterized by a visible dark line at the occlusal margin that could be stained by a probe, indicating marginal breakdown. Conversely, the bioactive restorations maintained their marginal integrity, likely due to the continuous formation of a hydroxyapatite-like layer at the tooth-restoration interface. No statistically significant differences were observed between the two groups for color match, surface texture, or anatomical form ($p > 0.05$ for all). Both groups maintained Alfa or Bravo ratings in over 92% of restorations for these parameters. A minor exception was observed in surface texture, where 6% of restorations in both groups exhibited slight superficial staining at the margins, which was easily removed by professional polishing and did not warrant a Charlie rating.

Patient-reported outcomes further supported the clinical findings, particularly regarding postoperative sensitivity. At the 24-hour follow-up, the incidence of postoperative sensitivity was significantly lower in the bioactive group (4.3%; 7 patients) compared to the control group (12.9%; 22 patients; $p = 0.021$). The severity of sensitivity, measured by the VAS, was also notably milder in the bioactive cohort. A concrete example of this difference was observed in a patient who received contralateral Class II restorations: the nano-hybrid restoration on tooth 37 elicited a VAS score of 45 mm in response to cold air and water stimuli, requiring the temporary avoidance of cold foods for five days, whereas the bioactive restoration on tooth 47 in the same patient yielded a VAS score of 0 mm, with the patient reporting no discomfort whatsoever. By the 1-week recall, postoperative sensitivity had resolved in all but two patients in the control group, who exhibited

lingering sensitivity and were monitored for potential pulpitis. Aesthetic satisfaction, evaluated using a 5-point Likert scale, was high in both groups, with 94% of patients in the bioactive group and 91% in the control group reporting being "satisfied" or "highly satisfied" with the overall appearance of their restorations at the 5-year mark, confirming that the therapeutic benefits of the bioactive glass did not compromise the aesthetic outcomes of the posterior restorations.

5 Conclusion

The present 5-year randomized controlled trial demonstrates that bioactive glass-containing composites offer superior long-term clinical performance compared to conventional nano-hybrid composites in posterior restorations. The significantly higher survival rate, coupled with a marked reduction in the incidence of secondary caries and improved marginal adaptation, confirms the clinical efficacy and reliability of these advanced materials. These findings strongly suggest that the continuous release of calcium, phosphate, and fluoride ions from the bioactive glass filler plays a crucial role in maintaining the structural integrity of the tooth-restoration interface. By promoting the formation of a hydroxyapatite-like layer and facilitating the remineralization of adjacent tooth structures, the material effectively mitigates the primary biological and mechanical causes of restoration failure. Consequently, bioactive composites represent a highly advantageous therapeutic option in daily clinical practice, particularly for patients with moderate to high caries risk, compromised oral hygiene, or those prone to marginal leakage, without compromising the aesthetic or functional requirements of posterior restorations.

Despite the robust randomized controlled design and a high recall rate of 91.4%, certain limitations of the study must be acknowledged. The research was conducted at a single university center, which may limit the generalizability of the findings to broader clinical settings with varying operator skill levels and patient demographics. Furthermore, while the evaluating examiners and statisticians were strictly blinded, the treating clinicians could not be blinded to the material used due to the distinct handling and rheological properties of the bioactive composite, which may have introduced a minor performance bias. Additionally, the study evaluated specific commercial brands of bioactive and nano-hybrid composites; therefore, the results may not be universally applicable to all materials within these broad categories, as formulations and filler technologies can vary significantly among manufacturers.

Future research should focus on multi-center trials with extended follow-up periods, such as 10 years, to assess the ultimate longevity, wear characteristics, and color stability of bioactive materials under prolonged functional loading. Further comparative studies are also warranted to evaluate different generations of bioactive composites, including bulk-fill variants, in high-stress-bearing areas, as well as their efficacy in pediatric populations and in the restoration of non-carious cervical lesions. Ultimately, continued investigation into the long-term biological interactions between bioactive materials and the oral environment will further elucidate the full clinical potential of this innovative material class and its role in shifting restorative dentistry toward a more preventive and regenerative paradigm.

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